

Decision Trees

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Verify the candidate scores for Fitting a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A stored fitted rule says spots ≥ 5 predicts spotted; otherwise healthy. Trace 4,5,6 and a missing count.
Policy: missing count requires review. Can test labels be used to choose this cutoff?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Rule: if spots ≥ 4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that training found the threshold?

6. Solve the given case

A supplied tree predicts spotted for spot_count ≥ 3 , otherwise healthy.

Trace 2,3,4 and state boundary.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A tree predicts fern when leaf_width_cm ≤ 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 11 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Draw the current Plant Classifier tree and annotate feature, threshold, prediction, and test evidence at each leaf.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Rule: if spots ≥ 4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that training found the threshold?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Name the parts and decisions

Identify the root, its two branches, leaf predictions and depth of the worked fitted tree. Which parts were selected using training labels?

My reasoning and evidence

13. W2 Verify the candidate scores

For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

My reasoning and evidence

14. W3 Separate fitting from prediction

A learner sees a blocked validation case with $\text{near}=0$ and changes that leaf to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

15. W4 Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

My reasoning and evidence

16. W5 Audit a tie and a claim

If near and tall make the same training error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Decision Trees

Trace a small fitted tree, including its exact boundary.

Toy stored tree: if visible_spots ≥ 2 predict spotted; otherwise healthy. Missing visible_spots requires review. Inputs L=1, M=2, N=None.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read first branch

L: $1 \geq 2$ is false, so predict healthy.



2. Read boundary

M: $2 \geq 2$ is true, so predict spotted.



3. Stop missing path

N lacks a count, so use review rather than inventing zero.

CHANGE ONE THING

The stored rule changes to $\text{visible_spots} > 2$.

Explain the changed result using the matching lesson.

Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace a small fitted tree, including its exact boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? N lacks a count, so use review rather than inventing zero.

1. Read boundary
2. Read first branch
3. Stop missing path

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Stop missing path N lacks a count, so use review rather than inventing zero. Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The stored rule changes to `visible_spots > 2`. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. M is now predicted healthy.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: M is now predicted healthy. Changing $\geq$ to $>$ moves the exact value 2 to the other branch.

Repair a missing building block

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using training labels only. Short practice: Can the candidate's training accuracy replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
